POLS 641
Introductory Political Analysis
Experiments in Political Science

Dr. Colin Kuehl

Northern Illinois University

November 10, 2026

Plan for Today

A broad introduction to experiments in Political Science

- Pop Quiz
- Course Logistics

- The Potential Outcomes Framework
- Reviewing the Logic of Experimental Design
- Experimental Models
- Problems that can arise in Experiments
- Types of Experiments and Examples
- Pre-registration
- Power Analysis

Course Logistics

Upcoming Schedule

- Experiments
- 11-12 Causal Inference & Experiments R code
- **11-17/19 Review & Midterm**
- 11-24 Thanksgiving Week - No Class
- 12-1 Mixed Methods
- 12- 3 Research Presentations
- 12-13 Final Project Due

Midterm

- November 13th - Full class period 2 1/2 hours - submitted by 6 PM
- Paper Portion
  - Multiple-choice, matching, short answer questions
  - Emphasis on understanding the logic of research and being able to "talk like a social scientist"
  - No notes, no computer, no friends
  - Submit before starting computer portion
  - Bring a pen . . . . and a positive attitude
- Computer portion
  - Mini-problem set format (Rmarkdown with questions and data)
  - You are free to use old notes, google, etc, but not your classmates, "the guru," or other humans
  - If you are a clueless code,r time may be an issue. If you're competent(which all of you are) it should not be.
- In total the midterm is worth 10% of your grade. This is equivalent to two problem sets and should reflect a similar level of effort.
- Learning this now will serve you throughout your career

Before class Tuesday

1. 4 questions for your classmates that could be on the midterm
2. 4 questions or clarifications on course concepts for me

Would a Reseach in Action article be useful?

Experiments in Political Science Research

- Experiments are becoming increasingly common in political science research.
  - Experimental articles have quintupled since the 1980s (Rogowski, 2016)

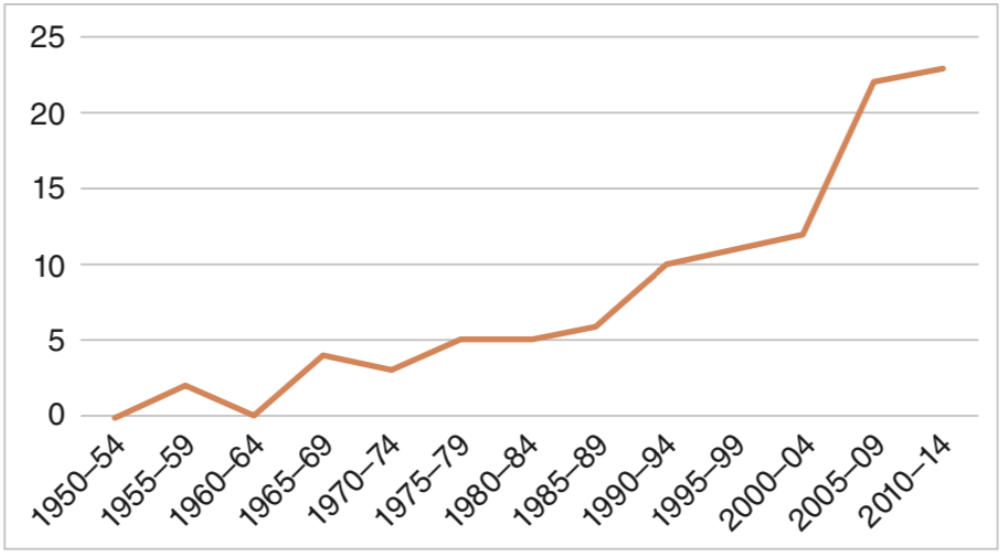


Figure 1 Experimental Articles in APSR, 1950–2014. (Note: Between 1906 and 1954, no experimental articles appeared.)

Why?

Why?

- What has changed in Political science?

Why?

- What has changed in Political science?
- What are the benefits of experiments over other designs?

Why?

- What has changed in Political science?
- What are the benefits of experiments over other designs?

Experiments in Political Science Research

- Randomized experiments deal with the endogeneity problem by creating exogeneity.

Experiments in Political Science Research

- Randomized experiments deal with the endogeneity problem by creating exogeneity.
- Q: What factors do we need to control for, as covariates, in an experiment?

Experiments in Political Science Research

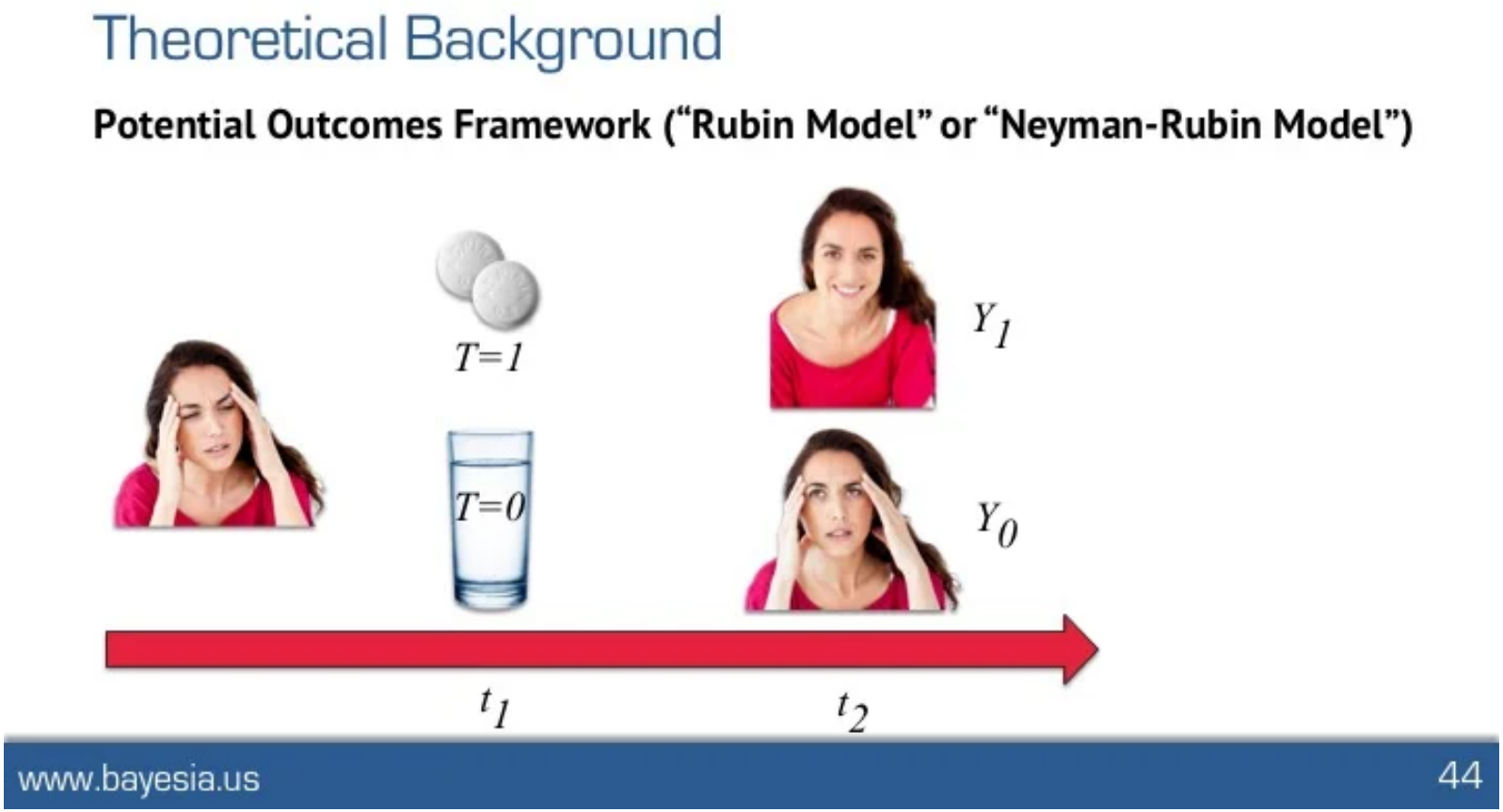
- Randomized experiments deal with the endogeneity problem by creating exogeneity.
- Q: What factors do we need to control for, as covariates, in an experiment?
  None! If randomization was done properly, and you have a large enough sample size, in expectation we should have balance across all the covariates.

Observational vs Experimental Studies

Observational vs Experimental Studies

Observational	**Experimental**
Researcher observes	Researcher intervenes (usually using random assignment)
Model-based inference	Design-based inference
Correlation	Causation
Externally Valid(usually)	Internally Valid (usually)
Often uses existing data	Cost and ethical data collection issues

Why Experiments: The Potential Outcomes Framework



The Potential Outcomes Framework

- There is no causation without manipulation
- **The fundamental problem of causal inference** at most one of the potential outcomes can be realized and thus observed.
- Causal inference is thus a **missing data problem**: given any treatment assigned to an individual unit, the potential outcome associated with any alternate treatment is missing.
- **Assignment Mechanism** How is it determined which units get which treatments or, equivalently, which potential outcomes are realized and which are not.

The Fundamental Problem of Causal Inference

The fundamental problem of causal inference is that, for every unit, we fail to observe the value that the outcome would have taken if the chosen level of the treatment had been different

We can only observe one potential outcome per unit (assuming we don't have access to a parallel universe)

The Fundamental Problem of Causal Inference

The fundamental problem of causal inference is that, for every unit, we fail to observe the value that the outcome would have taken if the chosen level of the treatment had been different

We can only observe one potential outcome per unit (assuming we don't have access to a parallel universe)

The search for causal inferences is therefore a search for assumptions under which we can infer the values of these unobserved counterfactual outcomes from observed data ie often we can think

about this as a missing data problem

The Potential Outcomes Framework

- Counterfactuals represent the outcomes that would occur under the opposite treatment assignment
- help us estimate the average treatment effect(ATE)

OLS Model for Experiments

$$Y_i = \beta_0 + \beta_1 D + \epsilon_i$$

- In experiments, we often denote the treatment variable as D. When D=1, the unit is treated, when D=0, the unit is a control.

OLS Model for Experiments

$$Y_i = \beta_0 + \beta_1 D + \epsilon_i$$

- In experiments, we often denote the treatment variable as D. When D=1, the unit is treated, when D=0, the unit is a control.
- We call β_1 the average treatment effect (ATE). Why?

OLS Model for Experiments

$$Y_i = \beta_0 + \beta_1 D + \epsilon_i$$

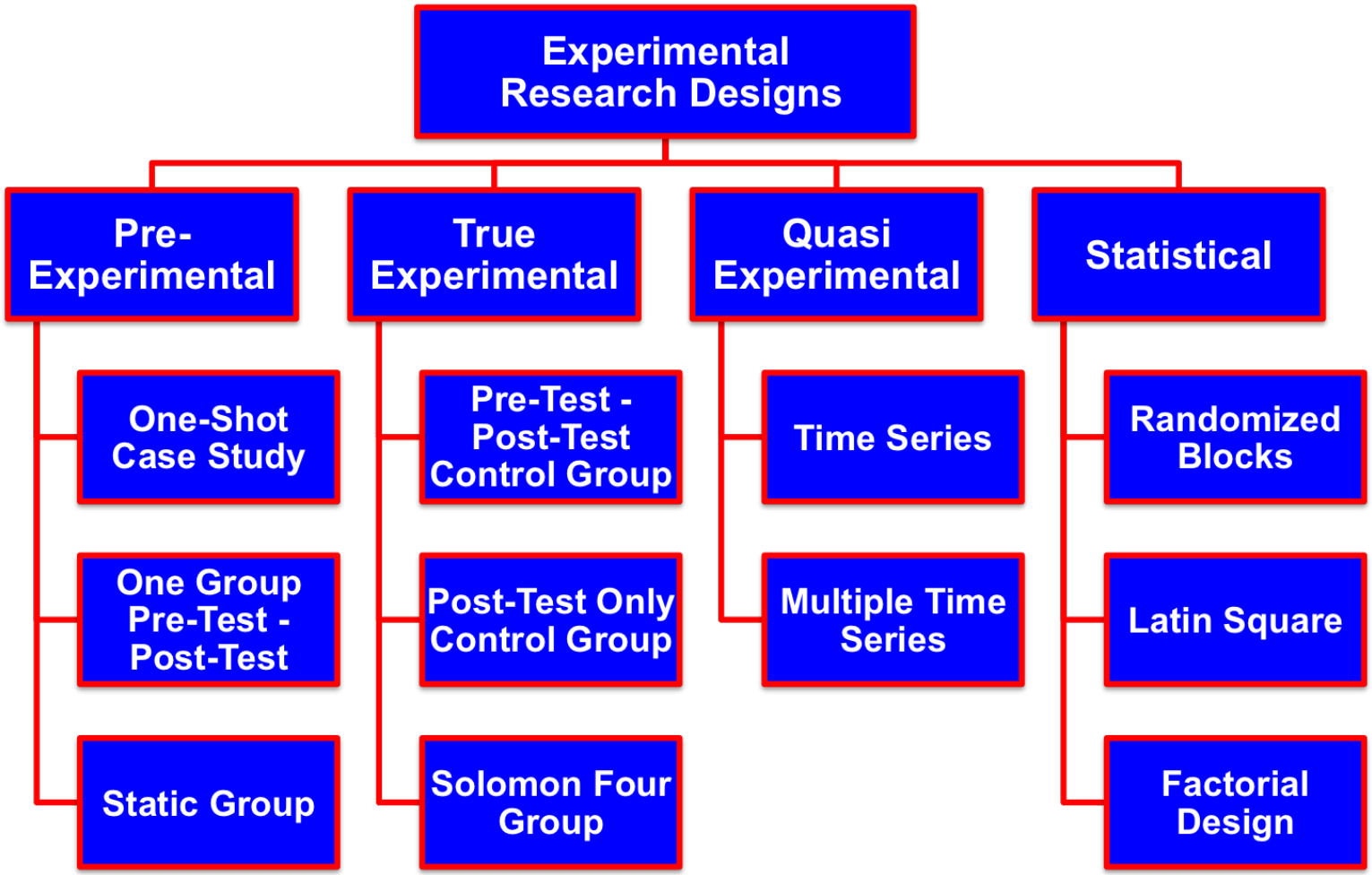
- In experiments, we often denote the treatment variable as D. When D=1, the unit is treated, when D=0, the unit is a control.
- We call β_1 the average treatment effect (ATE). Why? Since D can only take two values, a one unit shift in D is the effect of a unit moving from control (0) to treated (1). It's an average effect calculated across the full sample. This is the same as a dummy variable
- This is a relatively simple, bivariate regression. Why?

OLS Model for Experiments

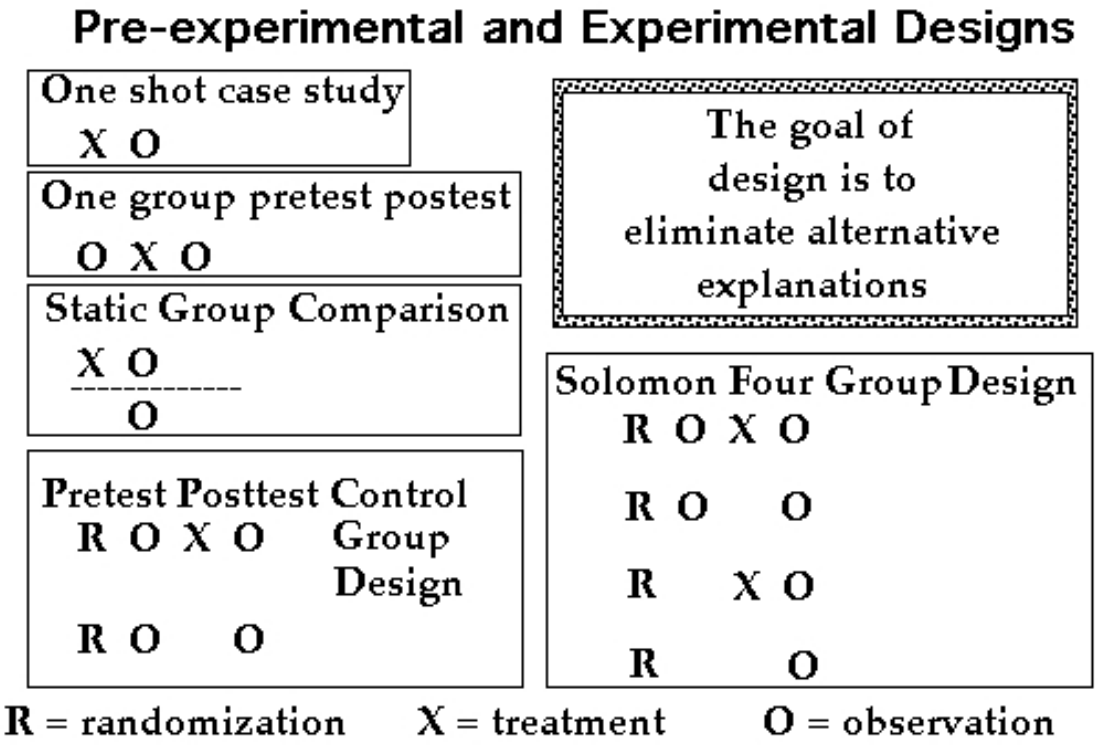
$$Y_i = \beta_0 + \beta_1 D + \epsilon_i$$

- In experiments, we often denote the treatment variable as D. When D=1, the unit is treated, when D=0, the unit is a control.
- We call β_1 the average treatment effect (ATE). Why? Since D can only take two values, a one unit shift in D is the effect of a unit moving from control (0) to treated (1). It's an average effect calculated across the full sample. This is the same as a dummy variable
- This is a relatively simple, bivariate regression. Why?
- Given proper balance in the covariates across the treated and control groups, we don't need to control for any covariates. In expectation (as the sample size grows) they are balanced across the two groups.
- IE We don't need to include control variables

Experimental Designs



Experimental Designs



Problems that can arise in Experiments

- Although many issues are addressed with experiments, most prominently endogeneity, there are still some pitfalls with experiments. What are they? (Bailey Ch 10).

Problems that can arise in Experiments

- Although many issues are addressed with experiments, most prominently endogeneity, there are still some pitfalls with experiments. What are they? (Bailey Ch 10).

- Attrition:

Problems that can arise in Experiments

- Although many issues are addressed with experiments, most prominently endogeneity, there are still some pitfalls with experiments. What are they? (Bailey Ch 10).
- Attrition: Units drop out of our experiment. Therefore, we never observed their outcome variable.
- Balance:

Problems that can arise in Experiments

- Although many issues are addressed with experiments, most prominently endogeneity, there are still some pitfalls with experiments. What are they? (Bailey Ch 10).
- Attrition: Units drop out of our experiment. Therefore, we never observed their outcome variable.
- Balance: Do the covariates (control variables) have the same mean across the two groups? We expect balance, given proper random assignment and large sample sizes.
- Compliance:

Problems that can arise in Experiments

- Although many issues are addressed with experiments, most prominently endogeneity, there are still some pitfalls with experiments. What are they? (Bailey Ch 10).
- Attrition: Units drop out of our experiment. Therefore, we never observed their outcome variable.
- Balance: Do the covariates (control variables) have the same mean across the two groups? We expect balance, given proper random assignment and large sample sizes.
- Compliance: Whether units actually receive the treatment they were assigned to (control or treatment).

Attrition

- Attrition is when units drop out of our experiment. For this reason, we never see the result of the experiment for this unit. In other words, the outcome variable is censored.
- Potential causes of attrition:

Attrition

- Attrition is when units drop out of our experiment. For this reason, we never see the result of the experiment for this unit. In other words, the outcome variable is censored.
- Potential causes of attrition: frustration with experiment, busyness, moving, death
- We might expect that the longer the experiment lasts, the more likely we are to see attrition.

- What is the problem with attrition?

Attrition

- Attrition is when units drop out of our experiment. For this reason, we never see the result of the experiment for this unit. In other words, the outcome variable is censored.
- Potential causes of attrition: frustration with experiment, busyness, moving, death
- We might expect that the longer the experiment lasts, the more likely we are to see attrition.

- What is the problem with attrition? It's likely not random. Some part of our sample will be more likely to attrit than others. This undermines the exogeneity created through random assignment.

- Potential solutions to attrition?

Attrition

- Attrition is when units drop out of our experiment. For this reason, we never see the result of the experiment for this unit. In other words, the outcome variable is censored.
- Potential causes of attrition: frustration with experiment, busyness, moving, death
- We might expect that the longer the experiment lasts, the more likely we are to see attrition.

- What is the problem with attrition? It's likely not random. Some part of our sample will be more likely to attrit than others. This undermines the exogeneity created through random assignment.

- Potential solutions to attrition? 1) Trimming the data set to make the treated and control units look the same on observable covariates (remember: this might not address unobservables). 2) Modeling the attrition to control for it.

Balance

- Balance is when your treated and control units look the same on other variables (covariates.)

Balance

- Balance is when your treated and control units look the same on other variables (covariates.)
- You can check this with a balance table: a table that describes the mean and variance for relevant control variables.

Balance

- Balance is when your treated and control units look the same on other variables (covariates.)
- You can check this with a balance table: a table that describes the mean and variance for relevant control variables.
- If your experiment was properly randomly assigned, and you had a large enough sample size (e.g. GG&L experiment) this shouldn't be a problem.

Balance

- Balance is when your treated and control units look the same on other variables (covariates.)
- You can check this with a balance table: a table that describes the mean and variance for relevant control variables.
- If your experiment was properly randomly assigned, and you had a large enough sample size (e.g. GG&L experiment) this shouldn't be a problem.

- Blocking:

Balance

- Balance is when your treated and control units look the same on other variables (covariates.)
- You can check this with a balance table: a table that describes the mean and variance for relevant control variables.
- If your experiment was properly randomly assigned, and you had a large enough sample size (e.g. GG&L experiment) this shouldn't be a problem.

- Blocking: Designing your treatment and control units in advance to ensure you will end up with balance in your covariates, particularly important confounders. This is most important for small sample sizes.

Balance

TABLE 10.1 Balancing Tests for the Progresa Experiment: Differences of Means Tests Using OLS

Dependent variable	$\hat{\gamma}_0$	$\hat{\gamma}_1$	t stat ($\hat{\gamma}_1$)	p value ($\hat{\gamma}_1$)
1. Age (in years)	1.61	0.01	0.11	0.91
2. Male	0.49	0.02	1.69	0.09
3. Child was ill in last 4 weeks	0.32	0.01	0.29	0.77
4. Father's years of education	3.84	−0.04	0.03	0.98
5. Mother's years of education	3.83	−0.33	1.87	0.06
6. Father speaks Spanish	0.93	0.01	1.09	0.28
7. Mother speaks Spanish	0.92	0.02	0.77	0.44
8. Own house	0.91	0.01	0.73	0.47
9. House has electricity	0.71	−0.07	1.69	0.09
10. Hectares of land owned	0.79	0.02	0.59	0.55
11. Male daily wage rate (pesos)	31.22	−0.74	0.90	0.37
12. Female daily wage rate (pesos)	27.84	−0.58	0.69	0.49
Sample size:	7,825			

Results from 12 different OLS regressions in which the dependent variable is as listed at left. The coefficients are from the model $X_i = \gamma_0 + \gamma_1 Treatment_i + v_i$ *(see Equation 10.2).*

Compliance

- Compliance refers to whether experimental subjects actually take the treatment if they were assigned to (and don't if they weren't assigned to.)

Compliance

- Compliance refers to whether experimental subjects actually take the treatment if they were assigned to (and don't if they weren't assigned to.)
- How could this happen?

Compliance

- Compliance refers to whether experimental subjects actually take the treatment if they were assigned to (and don't if they weren't assigned to.)
- How could this happen?
  Groups sneak into the treated group (e.g. the job training program) or the decline to take the treatment (e.g. they are assigned to come to a class to learn about statistics, but sadly, do not show up thereby missing their one shot at greatness!)

- What's one solution if you find you have non-compliance?

Compliance

- Compliance refers to whether experimental subjects actually take the treatment if they were assigned to (and don't if they weren't assigned to.)
- How could this happen?
  Groups sneak into the treated group (e.g. the job training program) or the decline to take the treatment (e.g. they are assigned to come to a class to learn about statistics, but sadly, do not show up thereby missing their one shot at greatness!)

- What's one solution if you find you have non-compliance?
  Run an intention-to-treat (ITT) model – compare the groups' by assignment, regardless of whether they took the treatment.

Compliance

- Compliance refers to whether experimental subjects actually take the treatment if they were assigned to (and don't if they weren't assigned to.)
- How could this happen?
  Groups sneak into the treated group (e.g. the job training program) or the decline to take the treatment (e.g. they are assigned to come to a class to learn about statistics, but sadly, do not show up thereby missing their one shot at greatness!)

- What's one solution if you find you have non-compliance?
  Run an intention-to-treat (ITT) model – compare the groups' by assignment, regardless of whether they took the treatment.
- What should we expect this to do to our ATE?

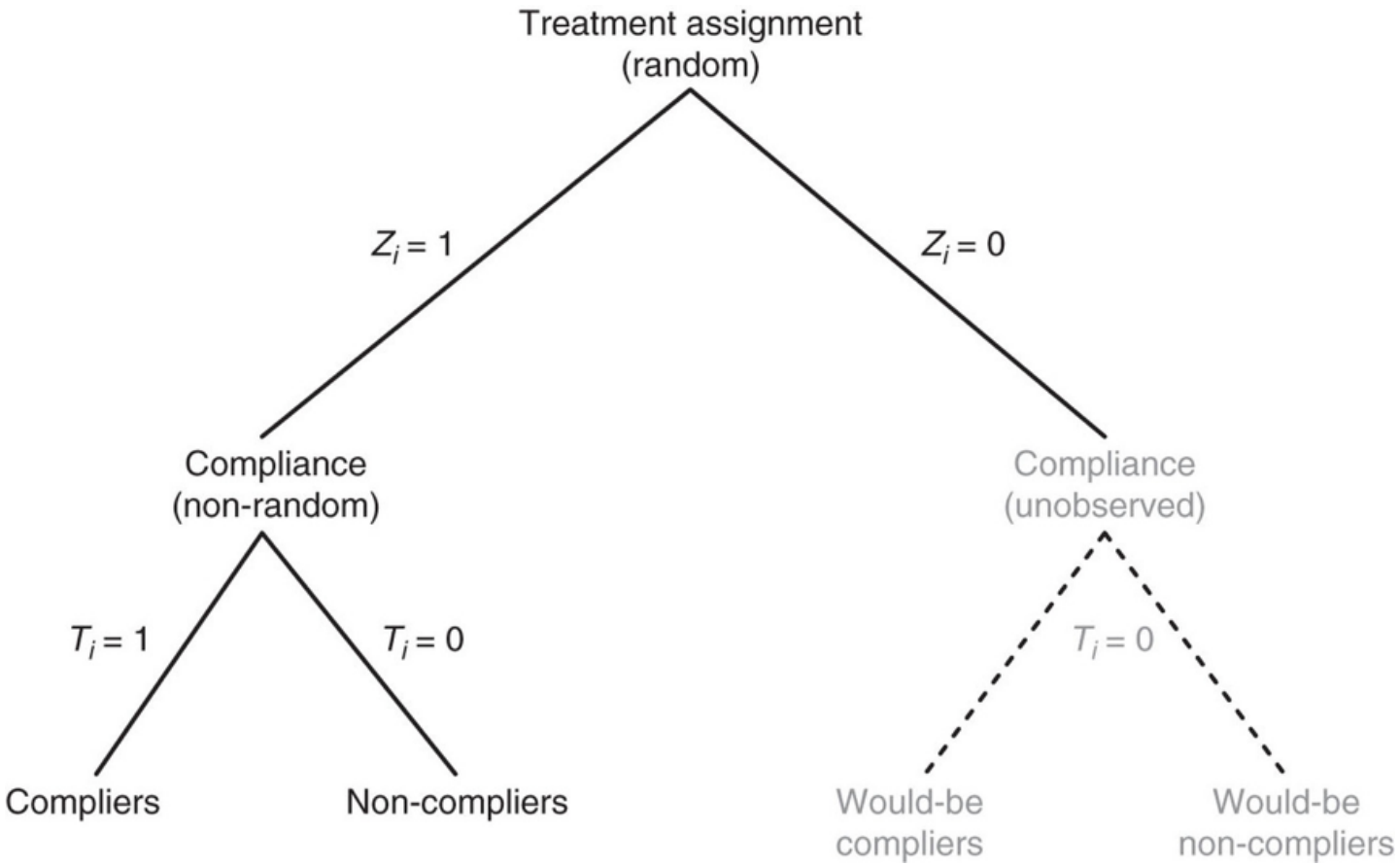
Compliance

- Compliance refers to whether experimental subjects actually take the treatment if they were assigned to (and don't if they weren't assigned to.)
- How could this happen?
  Groups sneak into the treated group (e.g. the job training program) or the decline to take the treatment (e.g. they are assigned to come to a class to learn about statistics, but sadly, do not show up thereby missing their one shot at greatness!)

- What's one solution if you find you have non-compliance?
  Run an intention-to-treat (ITT) model – compare the groups' by assignment, regardless of whether they took the treatment.
- What should we expect this to do to our ATE?
  Shrink it towards zero. As non-compliance grows, ATE gets closer to zero. This is like noise.

Never Takers: those that refuse treatment regardless of treatment/control assignment.
Always Takers: those that get the treatment even if they are assigned to the control group.
Defiers: Those that get the treatment when assigned to the control group and do not receive treatment when assigned to the treatment group. (these people violate an IV assumption referred to monotonicity)
Compliers: those that comply or receive treatment if assigned to a treatment group but do not recieve treatment when assigned to control group.

Compliance


Treatment assignment
(random)
$Z_i = 1$
$Z_i = 0$
Compliance
(non-random)
Compliance
(unobserved)
$T_i = 1$
$T_i = 0$
$T_i = 0$
Compliers
Non-compliers
Would-be
compliers
Would-be
non-compliers

Types of Experiments

- Traditional/Lab Experiments
- Survey Experiments
- Field Experiments
- Natural Experiments
- Not fully Experimental Experiments

Traditional Experiments

- Lab Experiments
  - simulations, computer tasks, etc.
  - Often used in political psychology
  - Completed in a “lab” where researcher has complete control

Traditional Experiments

- High Internal validity(high researcher control), but low external validity
- Participants tend to be WEIRD (participants are overwhelming Western, educated, and from industrialized, rich, and democratic countries)

Traditional Experiments: Gillam and Iyengar

Prime Suspects: The Influence of Local Television News on the Viewing Public

Franklin D. Gilliam, Jr. University of California, Los Angeles
Shanto Iyengar Stanford University

Local television news is the public's primary source of public affairs information. News stories about crime dominate local news programming because they meet the demand for "action news." The prevalence of this type of reporting has led to a crime narrative or "script" that includes two

Local television news is America's principal window on the world. Surveys of television viewing (e.g., Roper-Starch 1994), hours of daily programming (Papper and Gerhard 1999) and the actual share of the viewing audience captured by local newscasts (Hess 1991), all demonstrate the dominance of local news. In fact, people can watch live local news almost anytime—mornings, afternoons, evenings, prime time, and late night. As the amount of news time has increased, so too has competition between stations. The drive for audience ratings pushes local news or

Traditional Experiments: Survey Experiments

- Survey Experiments
  - Treatment in the form of text or images(ie read about this issue)
  - Efficiently run through survey software like Qualtrics. (NIU has free access)
  - Mturk Experiments are increasingly popular. Cheap and surprisingly representative
  - Participants have incentives move quickly / not pay attention
  - Some pushback from human subjects boards

Survey Experiments: Ardanaz, Murillo, and Pinto

Sensitivity to Issue Framing on Trade Policy Preferences: Evidence from a Survey Experiment

Martin Ardanaz, M. Victoria Murillo, and Pablo M. Pinto

Survey Experiments: Ardanaz, Murillo, and Pinto

- Group 1 (25 percent)—pro-trade introduction: "Some people believe that increasing trade with other nations creates jobs and allows you to buy goods and services at lower prices."
- Group 2 (25 percent)—anti-trade introduction: "Some people believe that increasing trade with other nations causes unemployment and hurts Argentine producers."
- Group 3 (25 percent)—both introductions.
- Group 4 (25 percent)—no introduction.

Survey Experiments: Ardanaz, Murillo, and Pinto

TABLE 2. *Percentages of respondents who favor increasing trade*

All respondents (N = 2,729)	71.31
Pro-trade introduction (N = 687)	69.43
Anti-trade introduction (N = 674)	67.95
Both introductions (N = 687)	69.29
No introduction (Group 4: N = 673)	78.46

Note: Table shows percentage who strongly agree and somewhat agree with the question.

The Planet's on F*cking Fire

Original Research

Science Communication
1–27
© The Author(s) 2023
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/10755470231153634
journals.sagepub.com/home/scx
SAGE

Exploring the Influence of Aggressive and Target-Framing Messages on Proenvironmental Behaviors

Shupei Yuan[1] and Colin Kuehl[1]

Abstract

As the tenor of climate change debates continues to intensify, it is critical to understand how shifts in style and target affect individuals' proenvironmental behaviors. Using two online experiments, we investigated how these changes can violate expectations in ways that affect persuasive power. The results suggest that an aggressive style is likely to expectancy violation, which negatively affects individuals' private- and public-sphere behavioral intentions. Differences in the framing of responsibility between individuals

Field Experiments

- a ”true” experiment conducted in a ”natural” setting
- Researcher has control over randomization - exogeneity
- Observe real world behaviors(as opposed to intentions, attitudes, etc) - greater external validity

Field Experiments

- Time consuming and expensive
- Often require partner organizations

Lacour and Green 2014 **infamous** example.
https://fivethirtyeight.com/features/how-two-grad-students-uncovered-

michael-lacour-fraud-and-a-way-to-change-opinions-on-transgender-rights/

Fields Experiments: Water Conservation

How Managers Can Reduce Household Water Use Through Communication: A Field Experiment

Heather Hodges
Colin Kuehl
Sarah E. Anderson
Phillip J. Ehret
Cameron Brick

Abstract

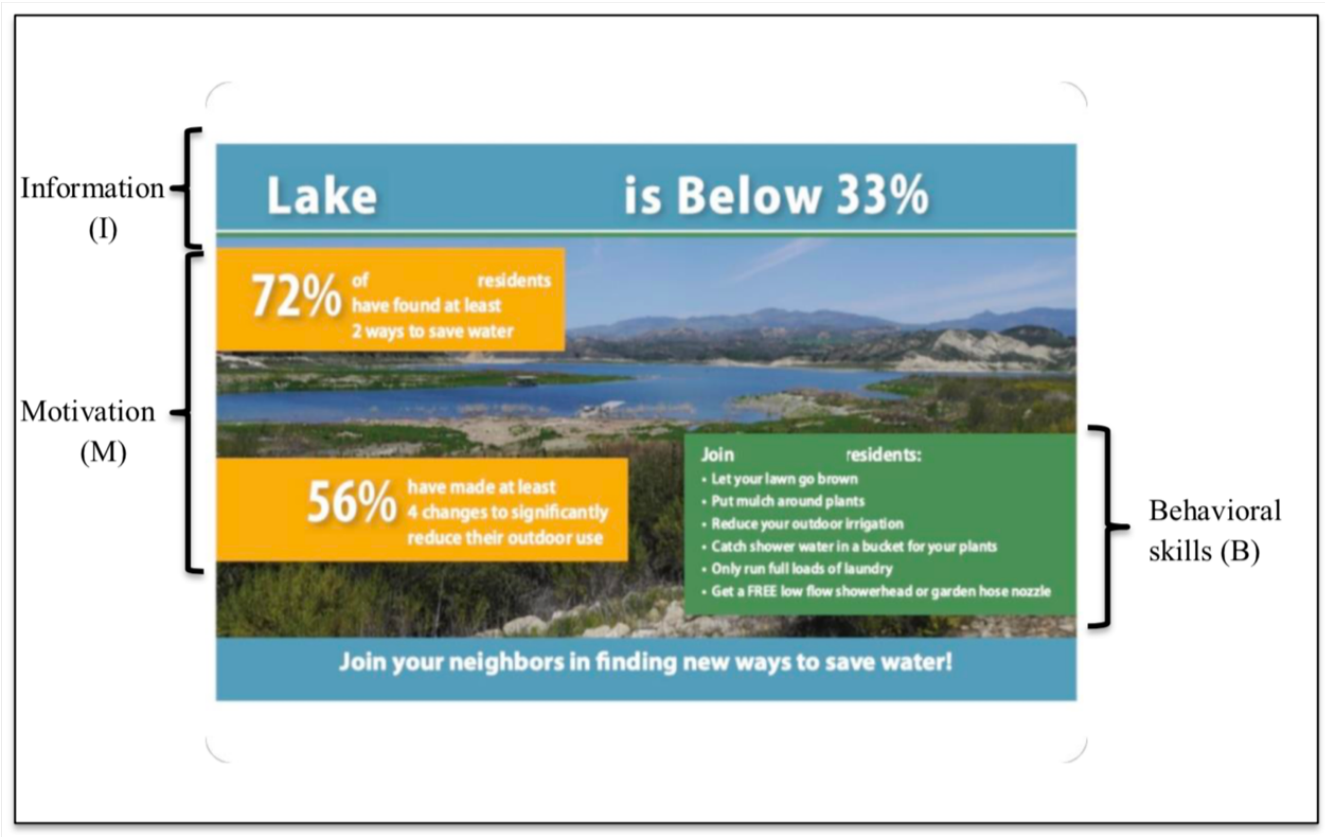
As populations increase and droughts intensify, water providers are using tools such as persuasive messaging to decrease residential water use. However, district-led messaging campaigns are rarely informed by psychological science, evaluated for effectiveness, or strategically disseminated. In collaboration with a water district, we report a field experiment among single-family households using persuasive messaging based on the information-motivation-behavioral skills model (IMB). We randomly assigned 10,000 households to receive different mailings and measured household water use. All messaging reduced water consumption relative to the control. On average, water use dropped 0.68 hundred cubic feet (HCF) (509 gallons) per household in the first month. Had all 10,000 single-family, occupied, non-agricultural residences been mailed the IMB messaging, more than five million gallons would have been saved in the first month. The effects declined but persisted for approximately three months and were three to six times greater in households with high water use (75th to 90th percentiles) relative to average water use. These findings suggest that combining message elements from the IMB model can reduce residential water use and that targeting high-use households is particularly cost-effective. *© 2020 by the Association for Public Policy Analysis and Management*

Fields Experiments: Water Conservation

Hypothesized differences in water use

	Control	IMB	IB	IM
Control				
IMB	IMB < Control			
IB	IB < Control	IMB < IB		
IM	IM < Control	IMB < IM	No prediction	

Fields Experiments: Water Conservation



Natural experiments

- Natural experiments occur when a researcher identifies a situation in which values of the independent variable have been determined by a random, or at least exogenous, process.
- We say treatment vs control is ”as if random”
- Hard to find, very effective when found

Natural Experiments Example: Posner 2004

American Political Science Review Vol. 98, No. 4 November 2004

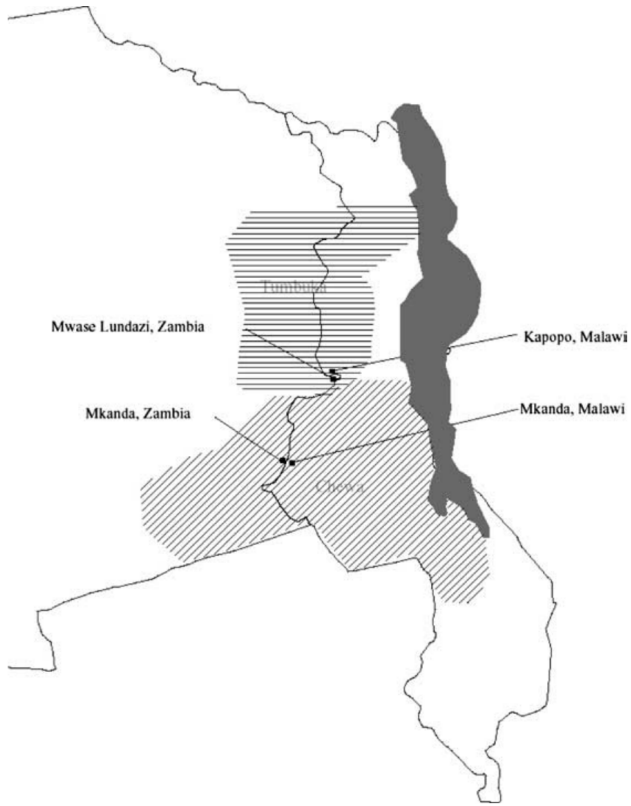
The Political Salience of Cultural Difference: Why Chewas and Tumbukas Are Allies in Zambia and Adversaries in Malawi

DANIEL N. POSNER *University of California, Los Angeles*

This paper explores the conditions under which cultural cleavages become politically salient. It does so by taking advantage of the natural experiment afforded by the division of the Chewa and Tumbuka peoples by the border between Zambia and Malawi. I document that, while the objective cultural differences between Chewas and Tumbukas on both sides of the border are identical, the political salience of the division between these communities is altogether different. I argue that this difference stems from the different sizes of the Chewa and Tumbuka communities in each country relative to each country's national political arena. In Malawi, Chewas and Tumbukas are each large groups vis-à-vis the country as a whole and, thus, serve as viable bases for political coalition-building. In Zambia, Chewas and Tumbukas are small relative to the country as a whole and, thus, not useful to mobilize as bases of political support. The analysis suggests that the political salience of a cultural cleavage depends not on the nature of the cleavage itself (since it is identical in both countries) but on the sizes of the groups it defines and whether or not they will be useful vehicles for political competition.

Natural Experiments Example: Posner 2004

FIGURE 1. Research Sites



Natural Experiments Example: Vietnam Draft

Deaths of Despair: Lessons from the Vietnam Draft Lottery

DALTON CONLEY and TIM JOHNSON

UNDERSTANDING THE CAUSES OF "DEATHS OF DESPAIR"[1]—that is, mortality due to suicide, drugs, and alcohol—is a fundamental challenge for public health research and a serious concern for the nation.[2] Studying that challenge, however, is difficult because the antecedents of such deaths are myriad and it is not possible to conduct a carefully controlled experiment to study the subject—or is it? In fact, 50 years ago this December 1, an experiment started that has helped illuminate the causes of deaths of despair and other public health challenges: the Vietnam Draft Lottery.

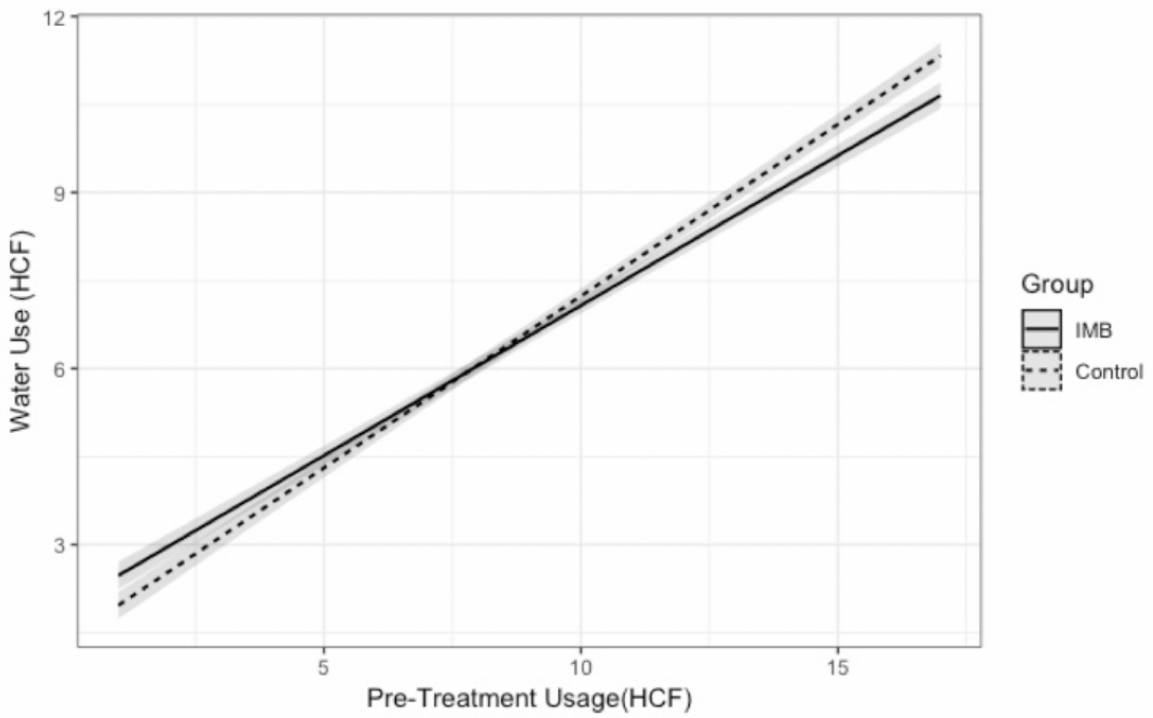
Other Forms

- Quasi-Experiments - Lacks randomness
  - Instrumental Variables
- Matching - Compare cases based on observable variables

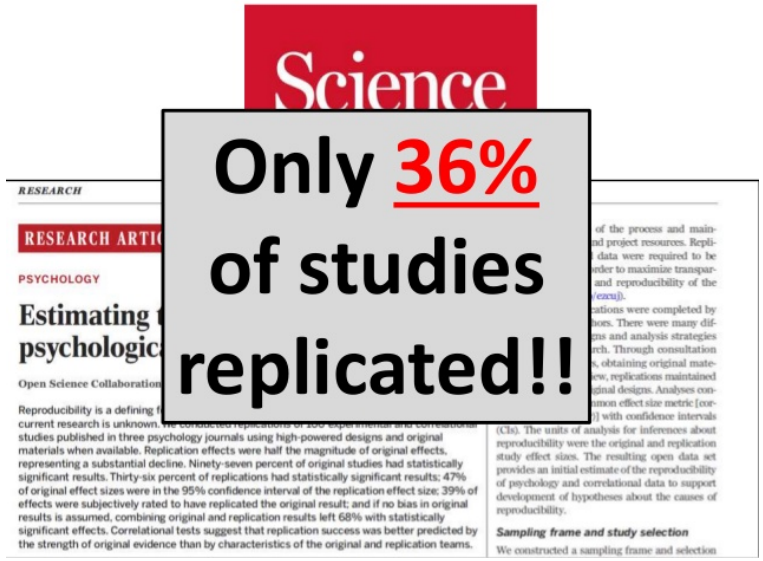
Key Term: Heterogeneous Treatment Effects

- Effect of the treatment is different for differing subjects
- Example: Expect to see greater conservation from water users with higher baseline use
- Modeled as an interaction term (treatment*baseline use)
- Often referred to as *Conditional Average Treatment Effect*(CATE)

Heterogeneous Treatment Effects



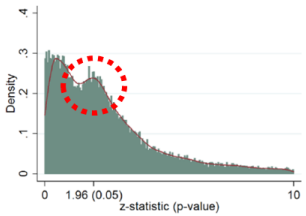
Psychology: The Replicability Crises and P-Hacking



Economics

Brodeur et al (AEJ:A, in press)

"Star Wars: The empirics strike back"

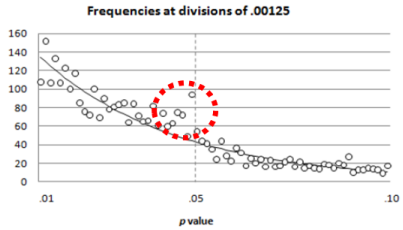


(b) De-rounded distribution of z-statistics.

Psychology

Masicampo Lalande (QJEP, 2012)

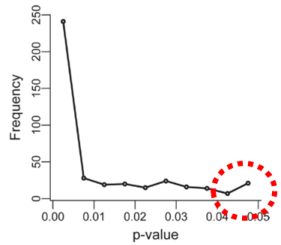
"A peculiar prevalence of p values just below .05"



Biology

Head et al (PLOS Biology 2015)

"Extent and Consequences of P-Hacking in Science"



The Replicability Crises and P-Hacking



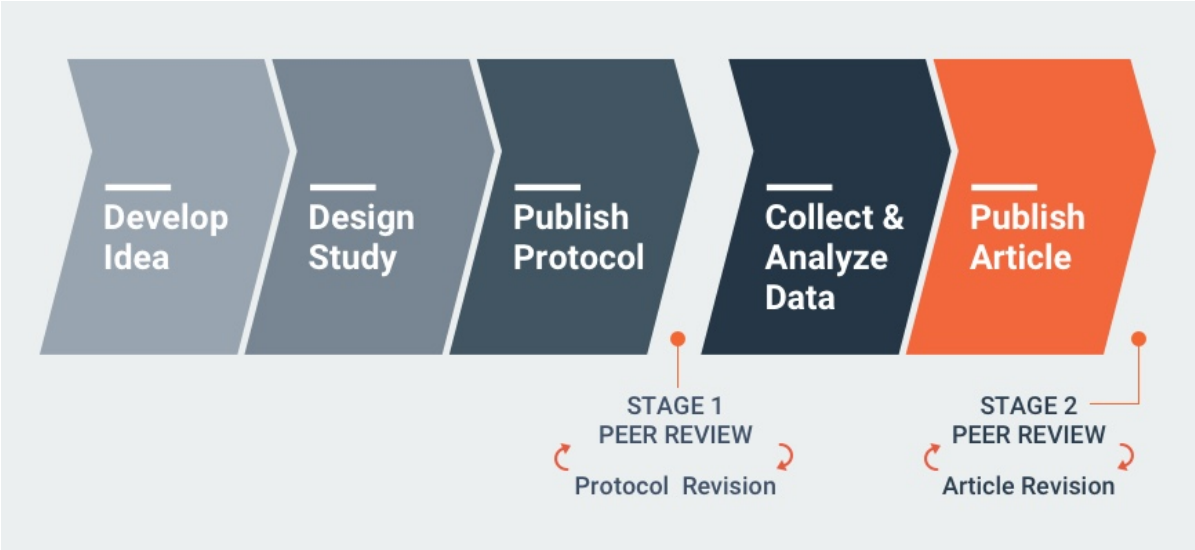
"If you don't reveal some insights soon, I'm going to be forced to slice, dice, and drill!"

Torturing the data til it speaks

Response: Pre-Registration

- Pre-Analysis Plan - Specifying the plan of your experiment before carrying it out
- Partial response to “Replicability Crisis” in psychology/social sciences
- Prevents exploratory hypothesis testing and p-hacking
- Registered Reports
  - Pre-registration is peer-reviewed and journals make provisional acceptance decision based on design
  - Decision is based on normal standards(contribution, novelty, methods, etc.), but without results

Pre-Analysis Plan and Registered Reports



Still more common in psychology and economics, but increasingly the norm in political science

Power Analysis

Power Analysis: Determining the minimum sample size to make statistically observations about the differences between groups.

Power is the ability to distinguish signal from noise.

Important Quantities:

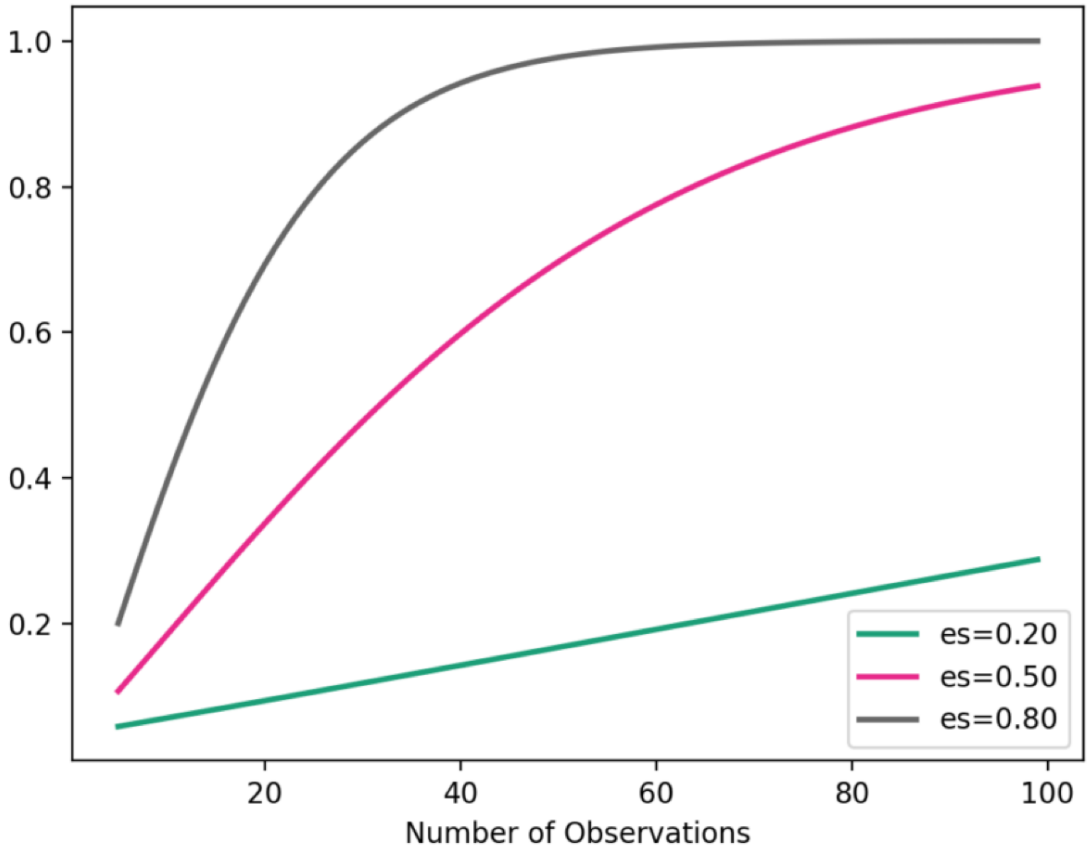
- Sample Size
- Effect Size: ATE
- Significance level: probability of finding effect that is not there (Type 1 error - false positive)
- Power: probability of finding an effect that is there (1-Type 2 error, false negative)

Power Analysis

Table 1. Types of Statistical Errors

	H_0 is actually:	
	True	**False**
Reject H_0	Type I error False positive	Correct
Accept H_0	Correct	Type II error False negative

Power Analysis



Review

- What are the pros of using experiments?
- What are the cons of using experiments?

- Bonus: Why have experiments only recently being used in political science?

Review: The ABCs of Experiments

- Attrition
- Balance
- Compliance

Review: Some Key Terms

- Random Assignment
- Attrition, Balance, Compliance
- Lab, Survey, Field, Natural, Quasi, experiments
- Others we skipped: ATE, ITT, CATE, PATE, SATE, ANOVA, ANCOVA
- Power Analysis
- Pre-Analysis Plan

Resources

- Gerber and Green (2012) *Field Experiments*
- Evidence in Governance and Politics - www.egap.org
- Center for Open Science - www.cos.io

Questions, Comments, Ideas?

Meme of the day



Current Events

